

A Hierarchical Electricity Consumption Forecasting Visualization System based on Multi-scale LSTM-KAN Model

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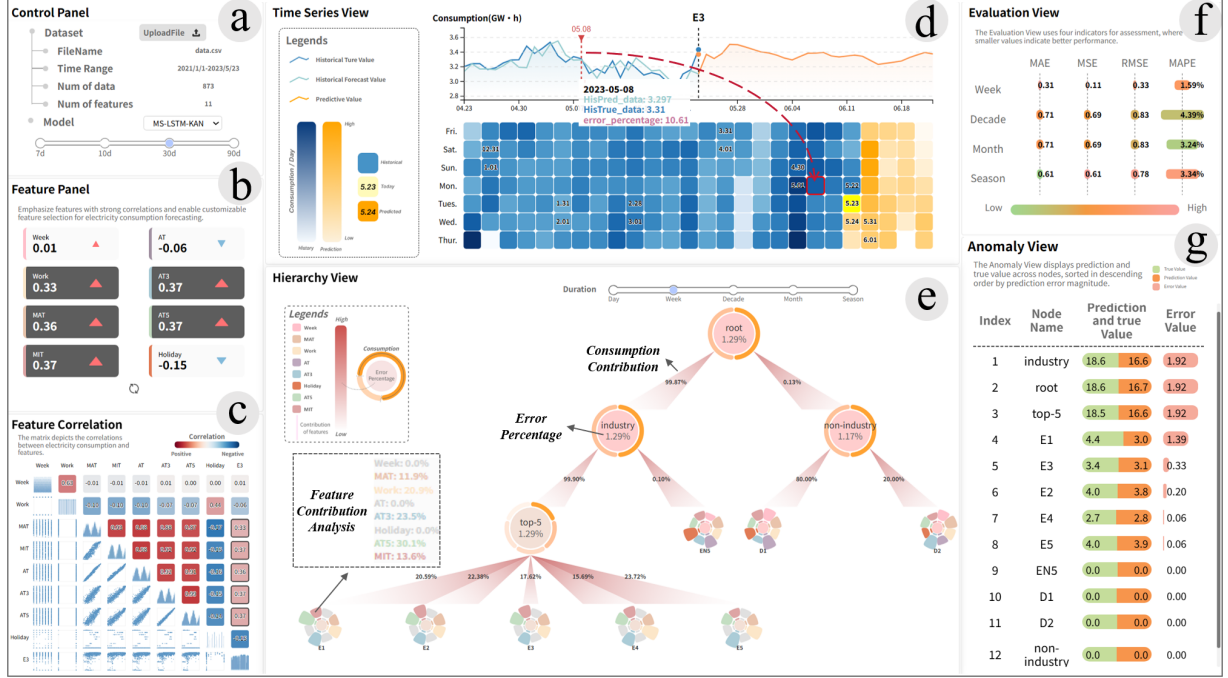


Figure 1: Visualization interface for electricity consumption forecasting: (a) a control panel for dataset selection, electricity forecasting model, and prediction day settings; (b) a feature panel containing eight relevant parameters for adjusting and regenerating predictions; (c) a feature correlation view displaying the correlation coefficients between target and features along with data distributions; (d) a time series view presenting historical predictions, true values, and future predictions via line and calendar charts; (e) a hierarchical structure view quantifying electricity contribution, error percentages, and leaf-node feature contributions to guide top-down predictive analysis; (f) an evaluation view for model performance assessment; (g) an anomaly view visualizing prediction error distribution.

ABSTRACT

Conventional electricity consumption forecasting methods typically integrate multiple factors, including historical power consumption data, natural environmental conditions, and socio-economic activities, to predict future electricity consumption within specific regions and time periods. However, the black-box nature of the prediction process and the lack of interpretability often hinder model optimization and limit improvements in prediction accuracy. In this paper, we propose a hierarchical electricity consumption forecasting visualization system. First, a bottom-up hierarchical forecasting architecture based on LSTM-KAN model is designed to achieve multi-scale predictions at fine-grained node levels in the underlying layers. Then, these predictions are progressively aggre-

gated in an interpretable manner to generate comprehensive consumption forecasts. Furthermore, we introduce a spatio-temporal coupled visual mapping method that presents prediction results and error distributions across different hierarchical levels and temporal scales through multiple dimensions. This approach supports interactive error backtracking and dynamic parameter adjustment, enabling visual validation and iterative optimization of prediction outcomes. Case studies using real-world urban electricity consumption data, quantitative analyses, and expert interviews demonstrate the effectiveness and practical utility of our method in optimizing electricity forecasting models and enhancing prediction accuracy.

Index Terms: Electricity consumption forecasting, visualization, LSTM-KAN, human-computer interaction.

1 INTRODUCTION

Electricity consumption forecasting, as a core component of smart grid operation and energy internet construction, plays a pivotal

role in supporting decision-making for power system planning, dispatch, safe and stable operation, and economic efficiency enhancement [3]. In practical power system operations, since electricity generation must dynamically match electricity demand, consumption forecasting becomes a critical decision-making basis for achieving peak shaving and valley filling, as well as avoiding redundant investments in power infrastructure [16]. It is particularly noteworthy that electricity consumption characteristics often exhibit significant nonlinear evolution patterns, influenced by the coupled effects of multidimensional dynamic factors such as climatic conditions, economic activities, consumer behavior and motivations, and energy policies [22, 31, 42]. Therefore, effectively constructing a multi-source feature fusion forecasting model to overcome the limitations of traditional methods in mining dynamic correlation features and capturing long-term dependencies has become a pressing scientific challenge in the field of electricity consumption forecasting.

Traditional electricity consumption forecasting typically relies on statistical models, such as Autoregressive (AR) [15] and Autoregressive Integrated Moving Average (ARIMA) [29], which predict future values through linear combinations of historical electricity consumption data. With technological advancements, the introduction of Machine Learning techniques [38], such as Random Forest (RF) [1] and Support Vector Machines (SVM) [33], has enabled the construction of multivariate feature spaces to capture the relationships between external factors (e.g. temperature, season) and electricity consumption. In recent years, Deep Learning models like Gated Recurrent Unit (GRU), Long Short-Term Memory (LSTM) [27], and Transformer [26] have gradually become mainstream. Their multi-layer neural network structures can effectively represent spatiotemporal nonlinear dynamic features in electricity data, significantly improving forecasting accuracy.

However, these methods still face considerable limitations in terms of forecasting accuracy and model interpretability. First, since the evolution of electricity consumption is influenced by exogenous factors (e.g., climate, holidays) and deeply intertwined with endogenous factors (e.g., regional industrial structure, economic development level), the aforementioned methods struggle to effectively integrate multi-source heterogeneous features and accurately and efficiently accomplish multivariate electricity consumption forecasting tasks (CH 1). Second, the "black-box" nature of forecasting models makes it difficult for users to analyze the dynamic relationships between input features and forecasting results, thereby hindering intuitive dissection of the forecasting process from different dimensions (CH 2). Finally, current forecasting methods often fail to precisely identify the sources of specific forecasting deviations, promptly detect potential issues, and perform reverse reasoning from error results to enable interactive optimization of forecasting outcomes (CH 3). Through close discussions with domain experts, a consensus was reached regarding these challenges: electricity consumption is inherently hierarchical, and its forecasting can be structured as a bottom-up multi-level forecasting system. Such a hierarchical forecasting framework supports fine-grained interpretability, making it easier to analyze consumption patterns and optimize forecasting results, which is of great significance for electricity consumption forecasting.

In this paper, we propose a hierarchical electricity consumption forecasting visualization system based on a multi-scale LSTM-KAN model. First, a hierarchical electricity consumption structure model is designed according to different consumption types. For each leaf node in the hierarchical structure, a multi-scale LSTM-KAN model is employed to capture electricity consumption features across different time scales, and the global forecasting result is obtained through bottom-up aggregation. Second, a spatiotemporal coupled visual mapping method is developed to dissect the forecasting process from macro to micro levels and present the

forecasting results. Finally, by integrating visualization techniques, the system interprets forecasting results based on feature contributions, analyzes electricity consumption trends, and supports interactive model optimization through user iteration. In summary, the contributions of this paper are as follows:

- A hierarchical multi-scale spatiotemporal electricity consumption forecasting method is proposed, where multi-scale LSTM-KAN models are used to forecast consumption for each leaf node in the hierarchical structure, followed by bottom-up aggregation to derive the total electricity consumption forecast.
- A hierarchical electricity consumption forecasting visualization system is developed to present forecasting results and error distributions across different levels and time scales, interpret results via feature contributions, and enable interactive model optimization.
- Case studies, quantitative analyses, and expert interviews based on real urban electricity consumption data further demonstrate the system's effectiveness in electricity consumption forecasting.

2 RELATED WORK

In this section, we mainly reviewed research in electricity consumption, time series forecasting, and visualization of time series data.

2.1 Research in Electricity Consumption

In recent years, with the accelerating global energy transition, the coordinated development of power system planning, operational management, and electricity consumption forecasting has provided crucial support for building new power systems and promoting sustainable energy development. Research on power system planning primarily focuses on maintaining the dynamic balance between electricity supply and demand. For instance, Urdaneta et al. [34] proposed a hybrid genetic algorithm based on successive linear programming, while Jin et al. [18] introduced a two-layer transmission expansion planning (TEP) method using voltage-source converter-based DC (VSC-DC). These planning methods enhance energy absorption capacity and economic efficiency, yet their effectiveness relies heavily on accurate electricity consumption forecasting. Merlin et al. [28] employed regression models to analyze factors influencing electricity consumption, revealing that economic growth, population, and power generation positively affect electricity demand. Ye et al. [39] used a fixed-effects panel model to examine the relationship between temperature and household hourly electricity consumption, further demonstrating the impact of climate on electricity consumption. These studies on influencing factors lay an important foundation for constructing high-precision electricity consumption forecasting models. Traditional statistical forecasting methods continue to evolve in power system modeling. Grey models (GM) and vector autoregression (VAR) [10] have shown effectiveness in small-sample and multivariate forecasting scenarios through algorithmic optimization. Gu et al. [14] developed an adaptive weighted least squares (AWLS) grey model to predict renewable energy generation trends, but it struggles to capture the complex fluctuations in renewable energy generation. He et al. [17] and Du et al. [12] proposed deep learning-based models, DDPG-GRU and MOWOA-Transformer, respectively, achieving high-precision electricity load forecasting with significantly reduced errors in single-step and multi-step predictions. While these studies have improved power system efficiency and reliability, the increasing heterogeneity of multi-source data in modern power systems and the volatility introduced by high-penetration renewable energy integration impose greater demands on forecasting models.

2.2 Time Series Forecasting

Time series forecasting, a core task in time series analysis, aims to infer future trends from historical observations, supporting dynamic decision-making in fields such as meteorology, finance, and industry. Traditional forecasting primarily focuses on linear modeling. Conejo et al. [9] combined wavelet transforms with ARIMA models to forecast electricity prices, achieving good results in the Spanish electricity market. However, such methods struggle to capture nonlinear patterns and complex dynamic changes. Through systematic feature engineering, machine learning methods can effectively extract nonlinear relationships in time series data, outperforming traditional linear models. Latifah et al. [21] applied random forests to predict forest fires, while Peng et al. [30] proposed an LS-SVM algorithm to find optimal classification hyperplanes for nonlinear problems. However, these methods face limitations in handling long-sequence dependencies.

Deep learning models have been widely adopted for time-series forecasting. Amalou et al. [2] and Xu et al. [37] applied RNNs and CNNs to energy consumption forecasting and bridge structural monitoring, respectively, improving prediction accuracy but suffering from vanishing gradient problems. Zheng et al. [40] introduced a GRU model with selective state updates to enhance accuracy. Kong et al. [19] proposed P-sLSTM, which segments time-series into patches to capture long-term dependencies. While Transformers effectively model long-range dependencies and outperform on large datasets, they require intensive computational resources and prolonged training durations [5]. However, they require substantial computational resources. Zhou et al. [41] proposed Informer, reducing computational complexity while maintaining performance in energy consumption and weather forecasting. Additionally, Chang et al. [8] proposed LLM4TS, a pre-trained large language model (LLM) framework for time-series forecasting, demonstrating strong generalization capabilities. Given data volume and computational constraints, we adopt LSTM as the base architecture with customized enhancements. Although existing approaches improve temporal prediction accuracy, their opaque decision-making and poor interpretability necessitate integrating visual analytics and interactive techniques for iterative model refinement.

2.3 Visualization of Time Series Data

Visualization of time series data is a key tool for understanding dynamic systems, converting multi-dimensional temporal data into intuitive visual representations to help analysts identify trends and refine strategies. Early approaches relied on basic charts such as line plots and scatter plots [36], which effectively display trends but struggle with large-scale, complex data. As research deepens, Deng et al. [11] introduced GeoChron, linking spatial context with evolutionary patterns to improve large-scale spatiotemporal data representation, though it struggles with adaptive times-tamp correlation. Walker et al. [35] proposed TimeNotes, reducing trend misidentification errors and enhancing multi-scale annotation capabilities. For feature representation, Bok et al. [4] developed PHP histograms, enhancing parallel coordinate plots (PCPs) with color-coded stacked histograms for intuitive attribute relationship inspection. Cakmak et al. [7] designed a recursive temporal summarization method, abstracting dynamic graph sequences into compact snapshots for periodic pattern detection in social networks. Shi et al. [32] introduced RankExplorer, extending ThemeRiver with color bars and dynamic symbols to reveal ranking fluctuations. Braun et al. [6] proposed magnitude horizon plots and logarithmic line charts for cross-magnitude data, improving trend detection accuracy. Devin et al. [20] introduced Aardvark, a composite visualization system combining tree-first, time-series-first, and image-first views for enhanced multi-perspective analysis. In contrast, we design an electricity consumption hierarchy that maps data across both temporal and spatial dimensions, employing interactive visual

analytics to interpret electricity consumption forecasting results.

3 TASK ANALYSIS AND SYSTEM OVERVIEW

In this section, we outline the analytical tasks identified through interviews with domain experts and introduce the pipeline of the proposed visual analytics system.

3.1 Task analysis

We conducted in-depth discussions and interviews with a power domain expert (E1), and an artificial intelligence specialist (E2). E1 is the director of the digital marketing department at a State Grid power supply company, specializing in electricity consumption forecasting and grid dispatch optimization, with extensive experience in consumer electricity behavior analysis. E2 is a university professor with nearly 20 years of experience in natural language processing (NLP) and a solid background in visualization design. During the early stages of our collaboration, we held weekly meetings with both experts to discuss their recent research projects in electricity consumption and time series forecasting. Both experts agreed that electricity consumption exhibits a hierarchical structure, and by classifying consumption based on industry sectors and geographical regions, forecasting accuracy and interpretability can be significantly improved. Furthermore, a hierarchical decomposition enables intuitive visual analysis of consumption patterns, facilitating better interpretation of forecasting results. Based on these discussions, we formulated a list of tasks as follows:

T1: Hierarchical Structure Construction. Electricity consumption inherently follows a hierarchical composition. For example, in a given city, large and extra-large industrial consumers significantly influence total electricity demand, while consumption across administrative districts is heavily affected by factors such as population density and economic development levels. To capture these variations, we classify urban electricity consumption along two primary dimensions: industry sector and geographical region. This hierarchical breakdown will enable fine-grained analysis of consumption patterns, improving the prediction of regional electricity trends.

T2: Bottom-Up Forecasting. After constructing a hierarchical structure for electricity consumption, a bottom-up forecasting approach is adopted. Electricity consumption is first predicted at the leaf nodes, i.e., the finest-grained user level (such as individual enterprises), using a multi-scale LSTM-KAN model. This model integrates industry-specific characteristics, climatic data, holidays, and other relevant factors to achieve accurate electricity consumption forecasting. Finally, the aggregates predictions step by step to generate a global electricity consumption forecasting result, reducing error accumulation and improving overall prediction accuracy, while also evaluating the model's forecasting performance.

T3: Pattern Visualization and Analysis. To facilitate intuitive understanding of electricity consumption patterns and insight into the forecasting process for decision-makers, the system presents prediction results from multiple dimensions and visualizes error distribution for further analysis. In the temporal dimension, it supports electricity consumption trends and forecast results at various time scales, including weekly, ten-day, monthly, and seasonal levels. In the spatial dimension, based on the hierarchical structure, it visualizes nodes representing different industries and users, with each node linked to its corresponding electricity consumption curve. Additionally, it quantifies the component contributions and overall impact of each node's electricity consumption.

T4: Interactive System Construction. To support in-depth analysis of forecast results and model optimization, a multi-dimensional interactive visualization system is developed. Based on the hierarchical structure, the system dynamically links and displays the corresponding forecast results and error distributions, enabling users to analyze the forecasting process layer by layer.

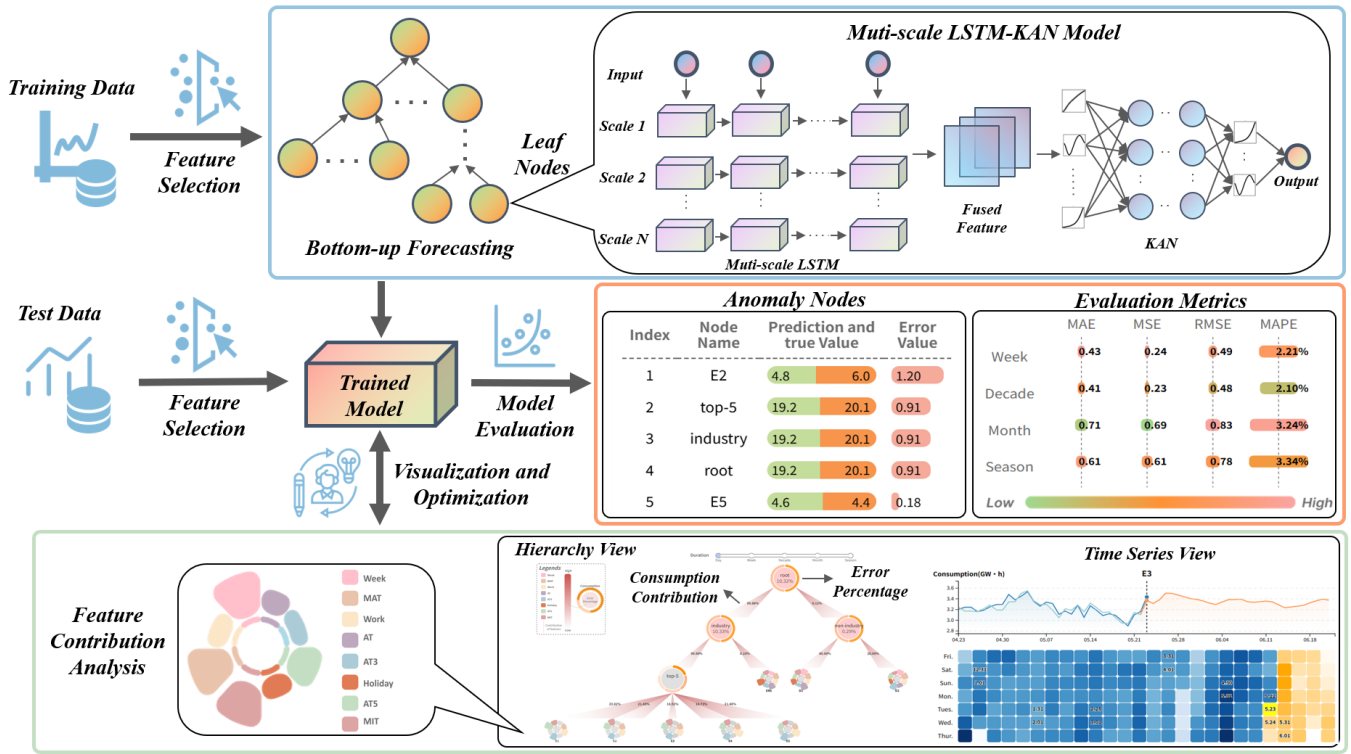


Figure 2: The pipeline of visual analytics system for hierarchical electricity consumption forecasting and interactive model optimization.

Through reverse inference of feature contributions, users can adjust feature parameters and iteratively optimize the model.

3.2 System overview

To accomplish the above tasks, we propose a hierarchical electricity consumption forecasting and visualization system that enables users to progressively explore the forecasting process from a macro to micro perspective and interactively optimize the model. The system pipeline is illustrated in Fig. 2. First, historical electricity consumption data and related influencing factors such as temperature and holidays are collected. After feature selection, a hierarchical electricity consumption structure is constructed based on industry and regional dimensions (T1) to capture local consumption behavior for more accurate forecasting. Next, using leaf nodes in the hierarchy as basic prediction units, a multi-scale LSTM-KAN model is applied to generate individual forecasts for each leaf node, which are then aggregated progressively to produce global consumption predictions (T2). Moreover, the results are visualized across temporal and spatial dimensions, with explainability achieved through analyzing feature contribution weights at leaf nodes, enabling users to examine predictions from macro to micro perspectives (T3). Finally, an interactive traceability mechanism is established, allowing model optimization through feature parameter adjustments informed by multidimensional error analysis and domain expertise (T4), culminating in the development of an interactive visual analytics system for electricity consumption forecasting.

4 METHOD

In this section, we introduce the proposed multi-scale spatio-temporal electricity forecasting and visualization framework based on LSTM-KAN. The framework aims to accurately forecast electricity consumption and analyze consumption patterns and trends from both temporal and spatial perspectives, enabling interpretation of forecasting results and optimization of the forecasting model.

4.1 Hierarchical Structure Construction

Electricity consumption data exhibits a hierarchical structure, which is essentially a tree that encodes spatial information of electricity consumption. We categorize electricity consumption by industry and administrative region, and further decompose it to finer granularity such as individual enterprises and local regions. Socio-economic factors such as economy, population, and electricity behavior characteristics are incorporated into the forecasting process to provide a structured basis for subsequent forecasting and visual analysis. To capture consumption patterns for specific regions, we construct a hybrid hierarchical structure combining industries and administrative regions. Let the electricity hierarchy be represented by a tree structure $T = (V, E)$, where V denotes the set of nodes, and E denotes the set of edges connecting parent and child nodes. According to industrial and non-industrial electricity consumption, the node set V is divided as $V = \{v_{\text{industry}}, v_{\text{non-industry}}\}$. Specifically, for v_{industry} , we divide electricity consumption into the top-5 largest enterprises and others. For $v_{\text{non-industry}}$, we categorize nodes by administrative region.

4.2 Bottom-Up Forecasting

We formulate the electricity consumption forecasting problem for each leaf node. For a specific node, the training samples are defined as $\{(X^i, y^i)\}_{i=1}^N$, where $X^i = [x_1^i, x_2^i, \dots, x_D^i] \in \mathbb{R}^{D \times T}$ represents the input features over T time steps and D dimensions (e.g., average temperature, holidays), $x_j^i \in \mathbb{R}^D$ represents the D -dimensional feature vector at the j -th time step, y^i represents the actual electricity consumption sequence, and N represents the number of samples in the training set. We train a multi-scale LSTM-KAN model, map the input features X^i through parameters Θ , and obtain the electricity consumption prediction value $\hat{y}^i$ by fitting the function $\hat{y}^i = f(X^i; \Theta)$. The system performs a bottom-up forecasting by recursively traversing the hierarchy from leaf to root nodes using

depth-first search (DFS), with each node's value computed as the sum of its children's predictions.

4.2.1 Feature Construction and Selection

Based on the electricity consumption characteristics, we construct time features such as weekdays, holidays, and climate features such as temperature. We also refer to the opinions of expert E1 and consider the thermal accumulation effect, using the average temperatures of the previous 3 days and 5 days as features. To further improve prediction accuracy, we use the Pearson correlation coefficient [24] to measure the strength of the relationship between electricity consumption values and features, selecting features that are more strongly related to electricity consumption values. The Pearson correlation coefficient ρ between feature f_j and electricity consumption value y is calculated using Eq. (1):

$$\rho = \frac{\sum_{i=1}^n (f_{ji} - \bar{f}_j)(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (f_{ji} - \bar{f}_j)^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

where f_{ji} and y_i represent the i -th elements of f_j and y , respectively, while $\bar{f}_j$ and $\bar{y}$ represent the mean values of f_j and y , respectively. The range of ρ is $[-1, 1]$, where 0 indicates no correlation, and ± 1 indicates perfect positive/negative correlation.

4.2.2 Multi-Scale LSTM

The proposed multi-scale LSTM-KAN model uses Long Short-Term Memory (LSTM) [13] as the backbone to address long-term dependencies, achieving more accurate time series prediction. LSTM controls the degree of discarding historical information through the forget gate, updates long-term memory by combining the input gate, forget gate, and candidate memory cell and finally controls the hidden state through the output gate to extract information and generate outputs. The structure of LSTM is shown in Eqs. (2) to (7):

$$f_t = \text{sigmoid}(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

$$i_t = \text{sigmoid}(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (4)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (5)$$

$$o_t = \text{sigmoid}(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

$$h_t = o_t \odot \tanh(C_t) \quad (7)$$

where x_t represents the input at the current time step, h_t and h_{t-1} represent the hidden states of the LSTM at the current time step and the previous time step, respectively. $\text{sigmoid}(\bullet)$ and $\tanh(\bullet)$ are two activation functions, applied to the forget gate f_t , input gate i_t , and output gate o_t . C_t represents the memory state at the current time step, $\tilde{C}_t$ represents the candidate memory cell, W_* and b_* represent weight matrices and bias vectors, and $\odot$ denotes element-wise multiplication.

To further improve the accuracy of long-term predictions and minimize the error accumulation caused by step-by-step predictions, we enhance the model's ability to capture multi-scale features using a multi-scale LSTM approach. We extract temporal features from different periods according to various time scales and finally complete feature fusion through fully connected layers, as shown in Eq. (8):

$$H^{\text{fused}} = W^{\text{fused}} \cdot \text{concat}(\{H^i\}_{i=1}^K) + b^{\text{fused}} \quad (8)$$

where H^{fused} represents the fused hidden state, K represents the number of different time scales, $H^i = [h_1^i, h_2^i, \dots, h_T^i]$ represents the hidden states of all time steps T under the i -th time scale, and W^{fused} and b^{fused} represent the weight matrix and bias vector, respectively.

4.2.3 Multi-Scale LSTM-KAN Model

KAN [25] is based on the Kolmogorov-Arnold theorem, using single-variable functions defined on the edges of the network as spline-based activation functions to replace linear weights. These functions are constructed as learnable activation functions, improving the accuracy of the network while reducing the number of model parameters. A KAN layer is defined as a matrix Φ composed of single-variable functions $\{\phi_{q,p}(\bullet)\}$, where $p = 1, 2, \dots, N_{in}$ and $q = 1, 2, \dots, N_{out}$, with N_{in} and N_{out} representing the number of inputs and outputs, respectively. $\phi_{i,j}(\bullet)$ denotes the trainable spline function. The L -layer KAN is expressed as Eq. (9):

$$y = \text{KAN}(x) = (\Phi_L \circ \Phi_{L-1} \circ \dots \circ \Phi_1)x \quad (9)$$

In the proposed multi-scale LSTM-KAN model, after inputting training samples, multi-scale LSTM feature extraction is performed to obtain the fused hidden state H^{fused} . This is then further processed through KAN via nonlinear transformations, ultimately outputting the prediction results. The model architecture is shown in Fig. 2 as the Multi-scale LSTM-KAN Model. Taking a two-layer KAN as an example, this is shown in Eq. (10):

$$\hat{y} = (\Phi_2 \circ \Phi_1)H^{\text{fused}} = \sum_{j=1}^K \phi_{1,j}^{(2)} \left(\sum_{i=1}^d \phi_{j,i}^{(1)} H^{\text{fused}} \right) \quad (10)$$

where the first-layer input dimension d is the dimension of H^{fused} , and the output dimension of the first layer matches the input dimension K of the second layer. The output dimension of the second layer is 1, representing the predicted output result. $\phi_{j,i}^{(1)}$ is the single-variable function from the first-layer output node j to the input node i . Similarly, $\phi_{1,j}^{(2)}$ represents the single-variable function for the second layer.

4.3 Spatio-Temporal Coupled Visual Mapping

The spatio-temporal coupled visual mapping method in this paper presents prediction results at different levels and time scales from a multidimensional perspective. By displaying the error distribution based on residuals between true values and predicted values, it helps analyze electricity consumption trends and explore prediction errors layer by layer to identify sources of prediction anomalies. Furthermore, for fine-grained nodes at the bottom layer, we use SHapley Additive exPlanation value (SHAP) [23] to explain the prediction process. The model's prediction value y_i for the i -th sample can be decomposed into the sum of SHAP values for each feature, as shown in Eq. (11). This allows us to calculate the contribution percentage of each feature to the model's prediction results, thereby explaining the prediction results of individual leaf nodes at different time dimensions and providing decision-making basis for adjusting feature parameters, achieving iterative optimization of the prediction model.

$$y_i = \phi_{i0} + \sum_{j=1}^M \phi_i(x_{ij}) \quad (11)$$

where x_{ij} represents the j -th feature of the i -th sample, $\phi_i(\bullet)$ denotes the calculation function of the SHAP value for the i -th sample's features, and ϕ_{i0} represents the baseline value of the i -th sample (usually the mean of the target variable for all samples in the test set).

5 SYSTEM INTERFACE

In this paper, we have developed a visual analytics system to interpret electricity consumption forecasting results, aiding users in analyzing and understanding the forecasting process from multiple dimensions. Fig. 1 showcases the visualization interface of the system, which includes a control panel and six main views.

5.1 Overview

In this system, the data file is first uploaded and the prediction model selected via the control panel Fig. 1 (a). Features relevant to each electricity consumption prediction unit are chosen based on the Pearson correlation coefficient, highlighted in the feature panel Fig. 1 (b). Subsequently, a bottom-up electricity consumption prediction is performed, with the forecasting outcomes and error distribution visualized through a spatiotemporal coupled visualization mapping method displayed in the time series view Fig. 1 (d) and hierarchy view Fig. 1 (e). The control panel allows for the selection of different models for electricity consumption prediction, and the evaluation view assesses the predictive performance of the models Fig. 1 (f). Upon completion of the prediction, users can examine the predicted values versus actual values of each node in the anomaly view Fig. 1 (g), and explore the prediction errors top-down along with the hierarchy view. By analyzing the petals of leaf nodes, the contribution of relevant features to the prediction results of that node can be determined, aiding user decision-making. Users can adjust feature parameters in the feature panel to accomplish iterative optimization of the model.

5.2 Feature Selection

As shown in Fig. 1 (c), the feature correlation matrix employs a multidimensional visualization approach to depict the statistical relationships between feature variables and the target variable, providing a basis for feature selection in predicting the target electricity consumption. The matrix consists of three distinct regions: the diagonal, the lower-left triangle, and the upper-right triangle. The diagonal region displays histograms that illustrate the distribution characteristics of each variable. The lower-left triangular region presents scatter plots that reveal bivariate relationship patterns among variables. The upper-right triangular region contains annotations of Pearson correlation coefficients, which quantify the strength of the correlations. These coefficients range $[-1, 1]$, where red indicates positive correlation and blue indicates negative correlation; the intensity of the color reflects the strength of the correlation. The last column of the matrix highlights the correlation coefficients between each feature and the target variable, with values closer to ± 1 indicating stronger correlation. These correlation coefficients correspond directly to the cards in the feature panel (Fig. 1 (b)), where each card is color-coded to represent the feature category and displays both the value and direction of the correlation ($\blacktriangle$ indicate positive correlations, while $\blacktriangledown$ indicate negative correlations). The size of each triangle visually represents the magnitude of the correlation. During prediction, the system automatically selects features based on the correlation analysis results, with selected features highlighted in dark gray. This enhances the input variable set for the forecasting model.

5.3 Spatio-Temporal Coupled Visual Mapping Method

Fig. 1 (d) and (e) present the prediction results of electricity consumption and error distribution from temporal and spatial perspectives, respectively. The Time Series View (Fig. 1 (d)) employs a combination of line charts and calendar maps to illustrate trends in electricity consumption. In these charts, blue, cyan, and orange lines represent historical actual values, historical predicted values, and future predicted values, respectively. Calendar maps visually reflect the periodic characteristics of electricity consumption through varying shades of color, with weekly labels and monthly separators enhancing the perception of temporal continuity. The Hierarchy View (Fig. 1 (e)) utilizes a tree structure to demonstrate the hierarchical relationship of electricity usage within a city, starting from the total electricity at the root node, progressively breaking down into industrial/non-industrial consumption, and finally into the finest granularity of enterprise/region-specific consumption. The node design employs differentiated visual encoding, al-

lowing multiple dimensions of information to be displayed within a single node without requiring view switching. Leaf nodes use petal-like elements around the periphery, where petal size represents feature contribution magnitude, and petal color corresponds to the features used in the forecasting process, enabling rapid anomaly tracing, as shown in Fig. 1 (e). Non-leaf nodes display the power distribution of constituent nodes via outer rings. All nodes include a red central circle, with its opacity/depth reflecting the magnitude of prediction error. Prediction processes employ a bottom-up aggregation mechanism, with nodes lighting up layer by layer as predictions progress, and final results are synchronously displayed in the line chart. Interactive design supports multi-dimensional exploration: the control panel can adjust the prediction duration, node clicks enable interactivity between the tree structure and line chart, and time dimensions can be freely switched between day, week, ten-day period, month, and season. Upon completion of predictions, the central ring of the root node displays overall error, allowing users to analyze error distribution by traversing down the tree structure. This design facilitates collaborative visualization of macro trends and micro features, providing power dispatch personnel with a multi-granularity analytical tool.

5.4 Model Evaluation and Iterative Optimization

Fig. 1 (f) evaluates the model's performance in predicting total electricity consumption across four dimensions: weekly, ten-day periods, monthly, and quarterly, using four error metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE). The smaller these metrics are, the more accurate the model's predictions are. Through the control panel, different model predictions can be selected to intuitively display the prediction outcomes. Detailed explanations of the four error metrics are provided in section 6.1 of this paper. In Fig. 1 (g), comparisons between predicted values and actual values for each node are shown, with green representing actual values, orange indicating predicted values, and pink depicting prediction error values. This visualization can be used in conjunction with the hierarchy view, allowing users to explore error distribution from top to bottom. When users drill down to leaf nodes, they can observe the contribution of relevant features to the prediction results through the petal representation on the leaf nodes, which further aids decision-making. Users can adjust feature parameters in the feature panel to iteratively optimize the model, thereby enhancing the prediction accuracy.

6 EVALUATION

Our system is developed using a Vue.js-based frontend, with D3.js employed for the design and rendering of visual views. MongoDB is utilized as the development database, while the backend is implemented using Python and PyTorch. The experiments are conducted on a GPU-equipped machine featuring a GeForce GTX 4050. The effectiveness and practicality of the system in electricity consumption forecasting are validated through quantitative analysis, two case studies, and interviews with two domain experts (E3 and E4).

6.1 Quantitative analysis

We utilize a real-world urban electricity consumption dataset, which records the electricity consumption of a certain city from January 1, 2021, to May 23, 2023. Seventy percent of the data are used for model training, while the remaining thirty percent is reserved for testing. To evaluate the performance of the proposed model, we adopt four widely used metrics: MAE, MSE, RMSE and MAPE, which are defined in Eqs. (12) to (15):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (12)$$

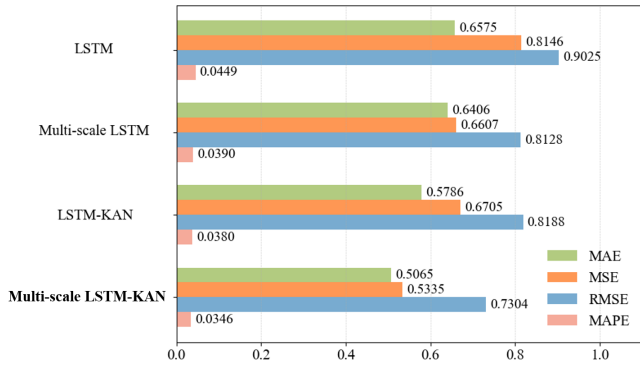


Figure 3: Comparative Analysis of Model Performance.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (13)$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (14)$$

$$MAPE = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (15)$$

where y_i and $\hat{y}_i$ represent the true value and predicted value of the i -th sample, respectively, and n denotes the number of samples in the test set.

As shown in Fig. 3, the experimental results demonstrate that the proposed multi-scale LSTM-KAN model performs well in the test set, with all error metrics outperforming those of the baseline LSTM model. Specifically, the low MAE value (0.5065) indicates that the predictions of the model are close to the ground truth. The low MSE (0.5335) and RMSE (0.7304) values suggest stable predictive performance. Furthermore, the relatively small MAPE (3.46%) further confirms the effectiveness of the model in practical applications. In addition, both the multi-scale LSTM and LSTM-KAN models exhibit lower error values compared to the standard LSTM, indicating that the multi-scale architecture and the integration of the KAN component contribute to improving the overall prediction accuracy of the network.

6.2 Case Study

In this section, we present two case studies involving experts E3 and E4 to illustrate how the proposed system can be used in practice to explore the electricity forecasting process and outcomes. Expert E3 has extensive experience in time-series data mining and analysis, while Expert E4 specializes in data visualization and human-computer interaction. In the first case, E3 utilizes real-world urban electricity consumption data to investigate how temporal and spatial scales can be integrated to forecast future electricity consumption. In the second case, E4 examines enterprise-level electricity forecasting results and explores whether the prediction performance can be further improved through interactive feature selection.

Case 1. Hierarchical Electricity Forecasting.

E3 conducted an analysis based on real electricity consumption data from a city. By default, the system employs the proposed MS-LSTM-KAN model to perform quarterly electricity forecasting. Upon receiving the forecasting command, the system automatically filters features for different leaf nodes, prioritizing those with stronger correlations, and subsequently aggregates the prediction results from the leaf nodes layer by layer to form the overall electricity consumption forecast for the city. To evaluate the performance of different models, E3 switched to the classical LSTM

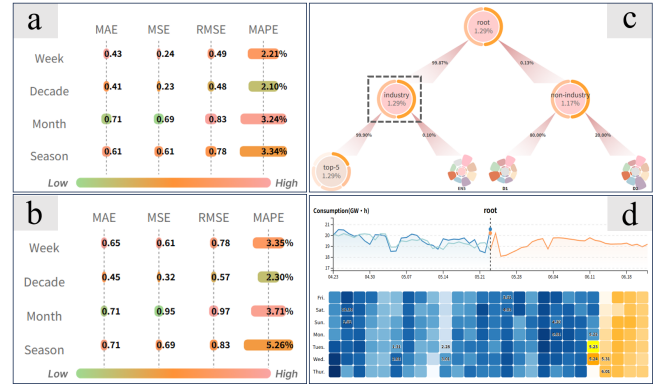


Figure 4: Hierarchical Electricity Forecasting: (a) and (b) evaluate the prediction performance of MS-LSTM-KAN and LSTM, respectively; (c) illustrates the hierarchical electricity forecasting process; (d) presents the overall electricity consumption trends and periodic patterns.

model and retrained it using the quarterly electricity data. A comparison of the results in the evaluation view (as shown in Fig. 4 (a) and (b)) revealed that the prediction curve generated by the MS-LSTM-KAN model closely aligned with the actual consumption curve (Fig. 4 (d)), and its prediction error was significantly lower than that of the LSTM model. Further exploration of the hierarchical view revealed that the industry node contributed more significantly to the overall electricity consumption, while the non-industry node had a relatively smaller impact. The contributions and reference features of lower-level nodes could also be clearly identified through the hierarchical structure (as illustrated in Fig. 4 (c)). E3 highly praised the model's performance, noting that its integration of multi-scale spatiotemporal features significantly enhanced the model's capability to learn complex electricity consumption patterns. Additionally, the incorporation of the KAN structure reduced the number of parameters without compromising accuracy, thereby improving computational efficiency. Moreover, E3 specifically highlighted the calendar heatmap (Fig. 4 (d)) for its ability to intuitively present historical electricity consumption patterns, which provides valuable insights for future model refinement.

Case 2. Interactive Model Optimization.

Building upon E3's research, E4 conducted an in-depth exploration of fine-grained optimization in prediction results. Using the system's time series view, E4 identified a significant deviation between the predicted value and historical data on April 30th (Fig. 5 (a)). To further investigate the cause of this discrepancy, E4 interacted with the line chart by clicking on April 30th to view detailed information. As shown in Fig. 5 (b), the industry node below the root in the hierarchical structure exhibited a darker central color, indicating a larger prediction error. E4 continued the exploration by drilling down into this node and ultimately pinpointed an anomalous prediction at the leaf node corresponding to Enterprise 2. The anomaly view (Fig. 5 (c)) also clearly visualized the substantial gap between the predicted and actual values for this enterprise on that day. Drawing on domain knowledge, E4 hypothesized that the anomaly might have resulted from a sharp drop in electricity consumption due to a holiday shutdown. To verify this, E4 accessed the feature analysis view for Enterprise 2 (Fig. 5 (d)) and discovered that the current prediction had not taken the holiday feature into account. Consequently, E4 manually added the holiday indicator as a critical feature and also incorporated historical electricity data from the same period in previous years to guide model retraining. As shown in Fig. 5 (e), the updated line chart and anomaly view indicate that the adjusted prediction closely aligns with the actual

values, with improved accuracy. Additionally, the petal diagram for Enterprise 2 demonstrates that the holiday feature made a meaningful contribution to the final prediction. The expert acknowledged the system’s powerful capability in anomaly detection and affirmed the practical value of human-in-the-loop optimization in predictive modeling scenarios.

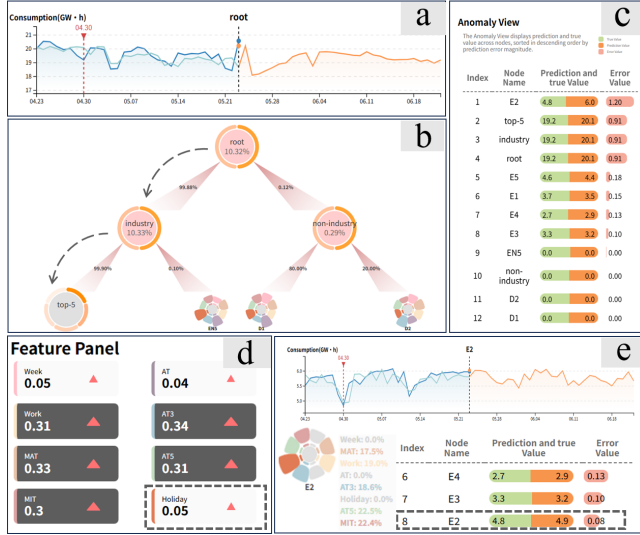


Figure 5: Interactive Model Optimization: (a) shows the prediction errors between forecasted and historical values; (b) illustrates the user’s top-down exploration process; (c) presents the anomaly distribution on the specific date; (d) indicates that the “Holiday” feature was not utilized in the prediction for Enterprise 2; (e) shows the updated line chart, prediction errors, and feature contributions after feature parameter adjustment.

6.3 Expert Interview

After domain experts (E1,E2,E3,E4) explored our experimental data set, we conducted semi-structured one-on-one interviews to collect their feedback on the usability, visual design and interaction of the system, which are summarized as follows.

System capability and effectiveness. The experts highly affirmed the prediction accuracy of the system. Expert E1 stated, “The hierarchical architecture based on regional and industrial classification effectively supports fine-grained exploration of complex power networks. Users can analyze electricity consumption processes layer by layer from a macro to a micro perspective, while simultaneously leveraging the statistical panel for real-time analysis. This design deeply integrates spatial dimensions with business logic, significantly enhancing analytical efficiency.” E2 added: “Hierarchical nodes and petal chart help intuitively visualize the contribution of electricity-related factors, enabling users to perform backward reasoning and better understand the forecasting process.” In addition, E3 noted, “This hierarchical electricity structure model aligns well with the characteristics of real-world power consumption hierarchies. Combined with bottom-up forecasting strategy, it significantly improves prediction accuracy. By incorporating electricity-related features at the leaf-node level, decision-makers can iteratively optimize the model through interactive analysis, further enhancing forecasting precision.” However, E3 also pointed out that “In practical power systems, unstructured data such as equipment inspection images and log texts exist but are not yet utilized in the current system. Incorporating such information in future may help explain sources of prediction error more effectively.”

Visual design and interactions. Experts unanimously agreed

that the system demonstrates superior performance in visual design and visualization methods, while also being user-friendly. Expert E1 stated: “The system presents forecasting results across different hierarchical levels and temporal scales, enabling dynamic cross-dimensional data correlation and progressive interpretation of the forecasting process. The correspondence between the hierarchical structure diagram and the line chart enhances the spatio-temporal coupling of the system.” E2 remarked, “The integration of multi-scale temporal analysis with hierarchical design makes the interface more visually engaging. The combination of calendar heatmap color encoding and line chart trend representation intuitively maps variations in electricity consumption intensity through color gradations. This design captures both short-term consumption fluctuations and long-term periodic patterns.” While affirming the effectiveness of the hierarchical view, E4 also provided valuable suggestions: “Considering the diversity in power structures across cities, as the number of nodes increases, the tree-based visualization can become visually cluttered. Future work should consider adaptive support for datasets with different structural characteristics to enhance the scalability of the system.”

6.4 Discussion

A series of quantitative analyses, case studies, and expert interviews indicate that our system can effectively support electricity consumption forecasting. However, there remain several unresolved challenges to be addressed in future work: (1) The current tree structure in hierarchy view is specifically designed for the power grid hierarchy of this city. In future, we will develop an interactive hierarchical structure construction feature to facilitate users in extending the system to diverse power grids (such as rural or other urban grids) and validate its generalizability. (2) The current design targets the electricity consumption forecasting scenario for this city. However, since LSTM models are widely applicable in time-series, its hierarchical interpretation framework can be extended to other scenarios such as load forecasting and renewable energy output prediction for model interpretation and optimization. This will require subsequent cross-domain validation. (3) The current system primarily utilizes structured numerical data such as historical electricity consumption and meteorological records. However, real-world power systems also involve unstructured data, including equipment inspection images and textual logs, which have not yet been fully leveraged. Incorporating such information in future could further enhance the model’s ability to explain prediction errors. (4) At present, we assess model performance using commonly used metrics such as MAE, which reflect accuracy of predictions to a certain extent. However, these metrics alone are insufficient for comprehensive evaluation. Our future goal is to incorporate additional information, such as business metrics from electricity dispatch systems, to further improve predictive accuracy.

7 CONCLUSION

In this paper, we proposed a bottom-up electricity consumption forecasting method based on the LSTM-KAN architecture, which performs multi-scale prediction at the fine-grained leaf-node level and aggregates the results to achieve an interpretable global electricity forecast. Furthermore, we designed a spatio-temporal coupled visual mapping method to present the prediction results and error distributions across different hierarchical levels and time scales. The system integrates rich human-computer interaction capabilities, allowing users to perform interactive backward analysis of prediction errors through feature attribution and error distribution insights, thereby enabling visual validation of results and iterative model refinement. Through extensive case studies and expert interviews, we have demonstrated the effectiveness of our approach in optimizing forecasting models and improving prediction accuracy, which is essential for applications in power planning and dispatch.

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